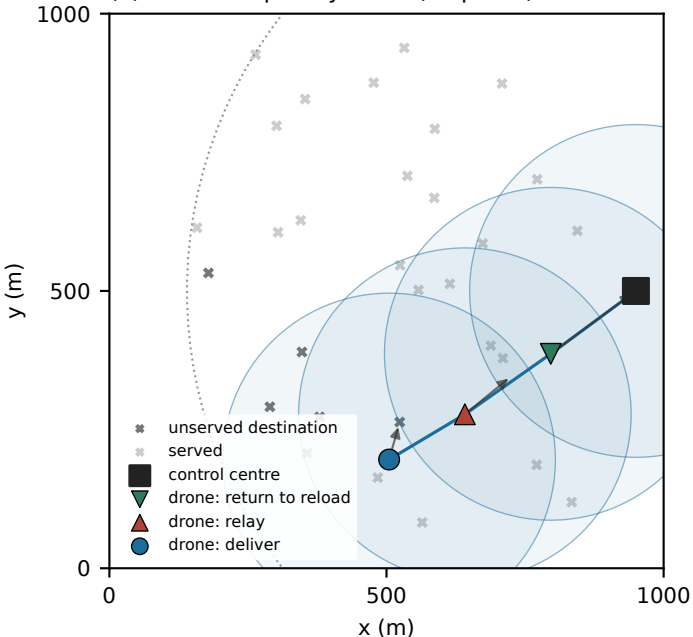
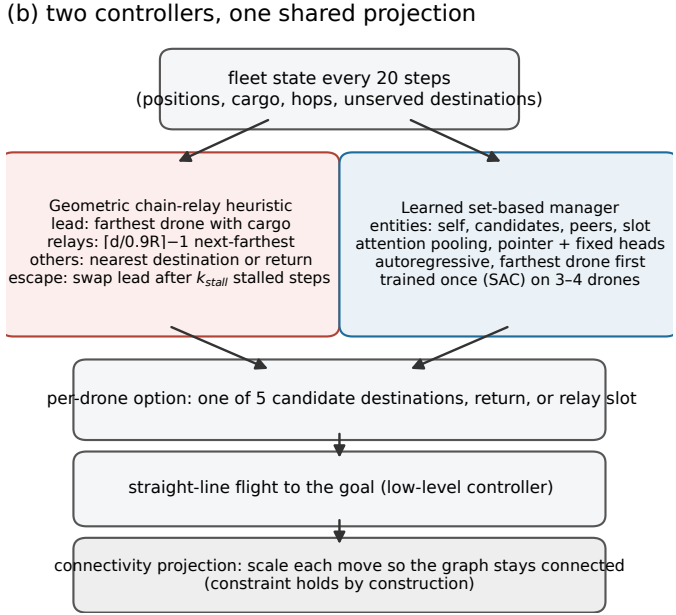

(a) a multi-hop relay chain (step 497)
1000
500
0
y (m)
0
500
1000
x (m)
unserved destination
served
control centre
drone: return to reload
drone: relay
drone: deliver


(b) two controllers, one shared projection
fleet state every 20 steps
(positions, cargo, hops, unserved destinations)
Geometric chain-relay heuristic
lead: farthest drone with cargo
relays: [d/0.9R]−1 next-farthest
others: nearest destination or return
escape: swap lead after k_{stall} stalled steps
Learned set-based manager
entities: self, candidates, peers, slot
attention pooling, pointer + fixed heads
autoregressive, farthest drone first
trained once (SAC) on 3–4 drones
per-drone option: one of 5 candidate destinations, return, or relay slot
straight-line flight to the goal (low-level controller)
connectivity projection: scale each move so the graph stays connected
(constraint holds by construction)